



BITS Pilani presentation

BITS Pilani
Pilani Campus

Faculty Name
Faculty Department



EEE ZG512 / SS ZG512, EMBEDDED SYSTEM DESIGN

Lecture No. 04



System - Design Requirements

- NRE cost (Nonrecurring Engineering cost): It represents the monetary cost that will not be happening again and again mainly the design cost
- Unit cost: Represents the monetary cost spent on manufacturing each copy excluding NRE cost.
- Performance: The execution time of the system or throughput of the system.
- Time-to-prototype: The time required to build a working model of the system, to refine the system's functionality.
- Power: The amount of power consumed by the system , which determines the lifetime of the battery.

User - Design Requirements

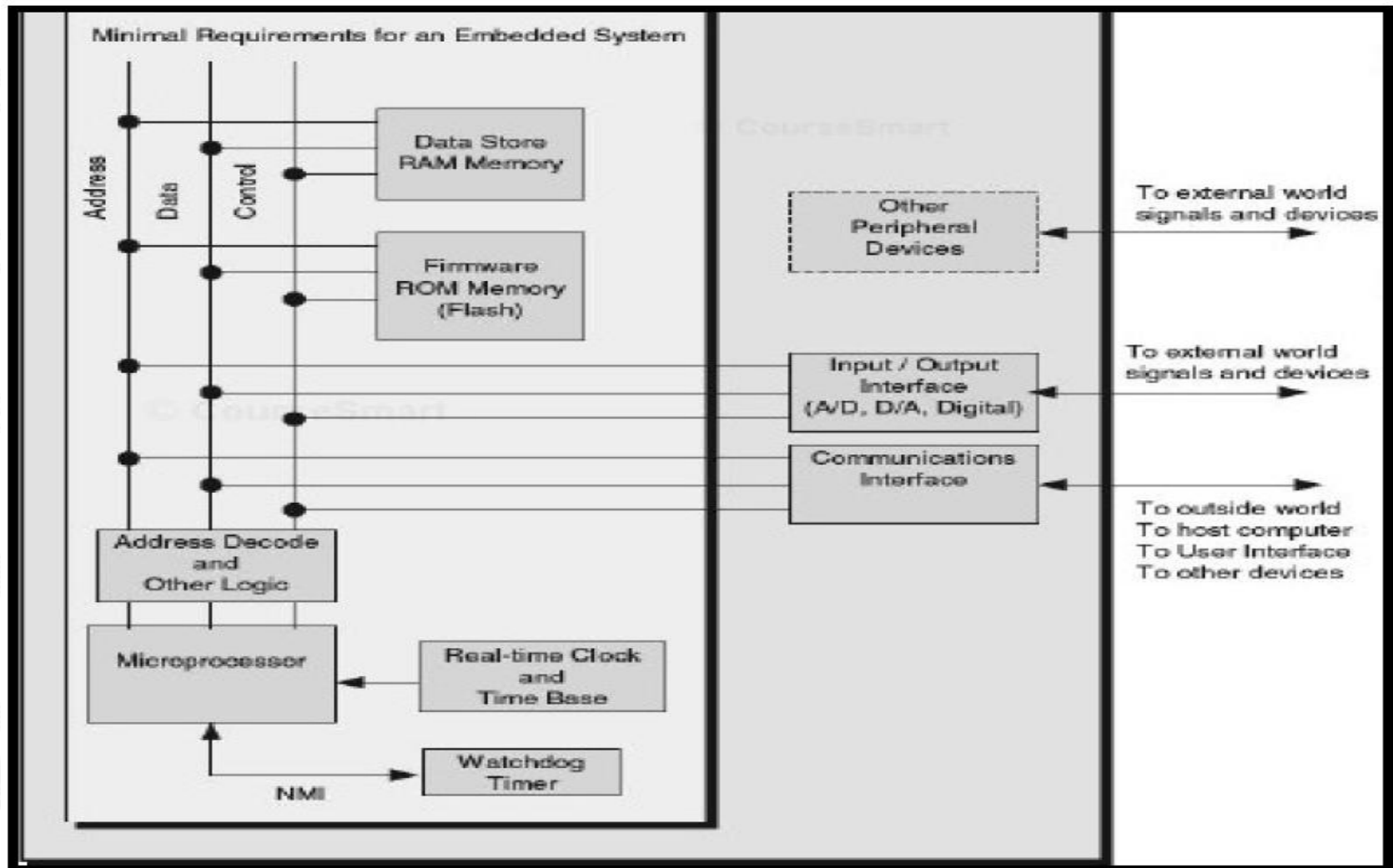
- Flexibility: Ability to change the functionality of the system without incurring heavy NRE cost
- Time-to-Market: The time required to develop the system to the point that it can be released and sold to customers.
- Maintainability: By the designers who did not do it originally. The ability to modify the product design after manufacturing.
- Correctness: Testing and verification
- Safety: The probability that the system will not cause harm.
- Size: The physical space required by the system, measured in bytes for software, and gates or transistors for hardware.

System Block Diagram

innovate

achieve

lead



Hardware Selection Criteria – Processing Unit



- Microcontrollers – Integrated CPU with memory, I/O, peripherals.
- Microprocessors – Standalone general-purpose CPU requiring external chips.
- DSP / IP – Integrated MAC (Multiply & Accumulate Unit) and FPU units.
- ASICs / ASIPs – Includes application-based circuit modules.
- FPGA – Designed flexibility to configure required modules for applications.
- Embedded Processors – Designed for specific operation.
- Common architectures: x86, ARM, AVR, PIC, 68k, MIPS, PowerPC

Memory Devices



- SRAM – Flip-flop based memory with Stable data and faster access rate.
- DRAM – Capacitor based memory with designed refresh cycle to retain data during operation.
- NVRAM – Non-volatile operation with RAM characteristics.
- EEPROM – Non-volatile operation that enables byte-wise access.
- Flash – Non-volatile that erases block-wise and allows byte wise write operations.
- Amount of memory depends on software needs.

I/O Interfaces



- Parallel Communication interfaces – PCI, SCSI
- Serial Communication interfaces – USART, SPI, I2C, USB
- Wireless Communication – Bluetooth, Wi-Fi, Zigbee, GSM
- Analog to Digital Converters (ADCs)
- Digital to Analog Converters (DACs)
- Timers & Counters



Power Supply



- Linear voltage regulators – Low noise with less efficient.
- Switching regulators – More efficient with switching noise.
- Supervisory circuits – Reset generation, brown-out protection.
- Backup supply – Batteries or capacitors



Clock Generation



- Crystal oscillators: High frequency with stable frequency.
- Tank Circuits (LC) : Medium frequency operations.
- RC Oscillators: Low frequency operations.
- Phase locked loops (PLLs): Configurable frequency generation.
- Real time clocks (RTCs) with backup.



END
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Session – 4
Thank You

